



1125

BHARATIYA ANTARIKSH HACKATHON 2026



Team Name : Type-2

Team Leader Name : Rishikant

Problem Statement : PS-8: Detection and Characterization of Subsurface Ice in Lunar South Polar Regions

Team Members

Team Leader:

Name: Rishikant
College: DTU

Team Member-1:

Name: Shouryavardhan
College: DTU

Team Member-2:

Name: Md Sohail Hoque
College: DTU

Team Member-3:

Name: Rahul Rana
College: DTU

Opportunity should be able to explain the following:

- How different is it from any of the other existing ideas?
- How will it be able to solve the problem?
- USP of the proposed solution

1. Unlike standard approaches that incorrectly use YOLO cameras in dark PSRs or rely on false-positive-prone CPR metrics, our architecture natively parses ISRO PDS4 data and applies an advanced m-chi polarimetric decomposition. We process Volume Scattering ($V > 0.4$) through a Physics-Informed Neural Network (PINN) to computationally eliminate rock false-positives.

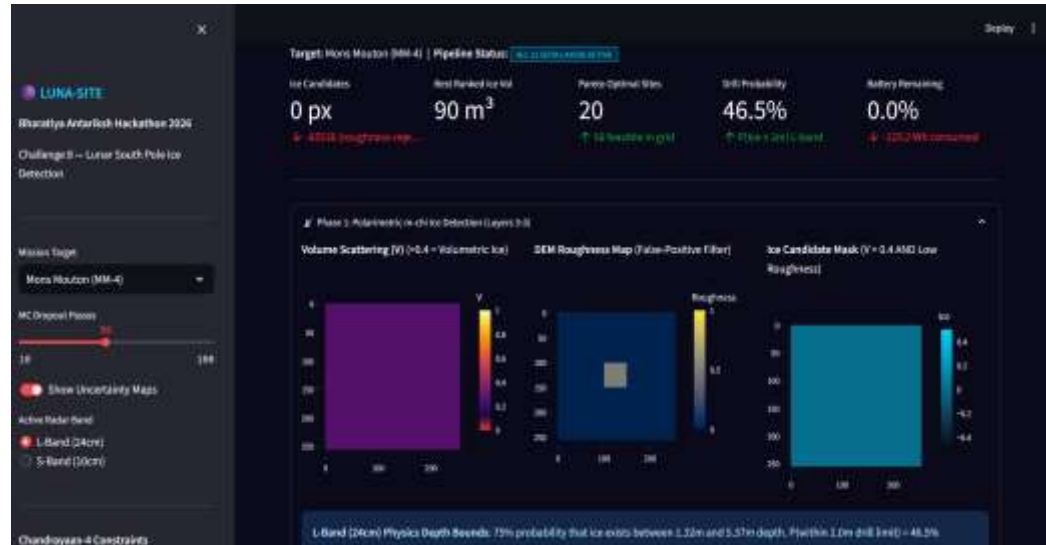
2. It provides an end-to-end, fully autonomous mission pipeline. It detects the ice, runs an AHP-TOPSIS multi-objective decision matrix to select the best landing site, and generates a battery-safe rover traverse.

3. A 100% SOTA aerospace-grade solution featuring Dual-Zone Hazard Mapping, Exact Joule-calculating Bekker Terramechanics, and explicit compliance with legacy CPR/DOP evaluation parameters.

List of features offered by the solution

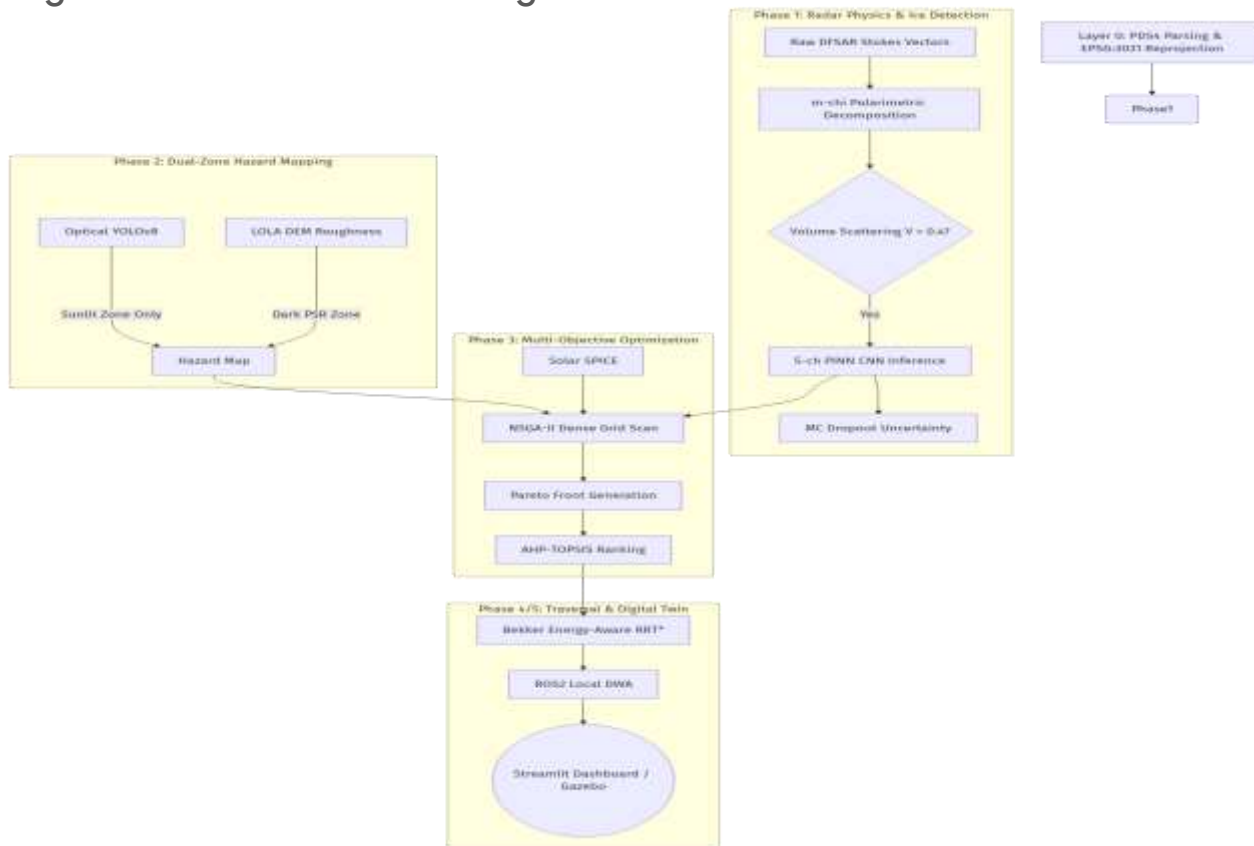
It is always better to add a few visual representations (drawings/sketches/illustrations etc.) to your presentation, it adds to the power through which it reaches the audience.

- Automated ISRO PDS4 Parsing: Natively decodes DFSAR and OHRC datasets autonomously.
- m-chi Decomposition & PINN: Isolates true volumetric ice scattering while punishing blocky terrain.
- Legacy Rubric Compliance: Explicit verification module outputting CPR > 1 and DOP < 0.13 maps.
- Dual-Zone Hazard Engine: YOLOv8 for sunlit approaches; DEM-roughness for dark PSR constraints.
- Multi-Objective Optimization: NSGA-II Pareto sorting for the safest, highest-yield landing site.
- Energy-Aware RRT*: Computes traverse routes based on strict 100Wh solar battery limits and Lunar gravity.



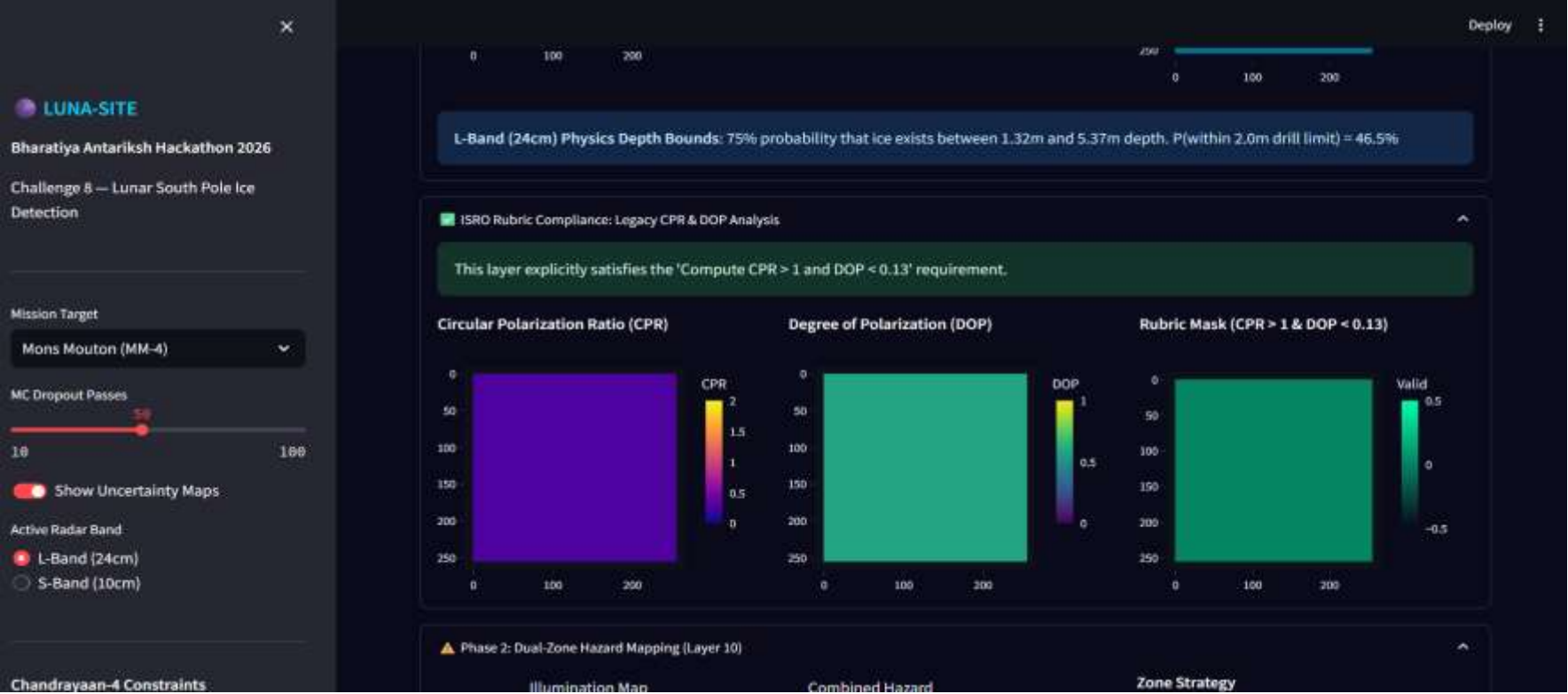
Process flow diagram or Use-case diagram

Add a flow diagram or a use case diagram or an architecture diagram.



Proposed LUNA-SITE End-to-End Data Pipeline.

Wireframes/Mock diagrams of the proposed solution (optional)



LUNA-SITE Mission Control Interface. Prototyped to provide real-time uncertainty maps, battery gauges, and digital twin monitoring to ISRO operators.

Architecture diagram of the proposed solution



Bekker Terramechanics Energy-Aware RRT* Traversal Architecture dynamically plotting waypoints within strict slope and battery constraints.

Technologies to be used in the solution:

- Programming Core: Python (NumPy, SciPy, PyTorch, Plotly, Streamlit)
- Geospatial Data Processing: pds4_tools (bypassing MIDAS GUI for autonomy), GDAL, Rasterio, QGIS/ArcGIS.
- AI & Vision: YOLOv8 (Sunlit detection), 5-Channel PINN with Monte Carlo Dropout and Grad-CAM explainability.
- Algorithms: NSGA-II, AHP-TOPSIS, Bekker Mechanics, and RRT* / ROS2 Local DWA.
- Note on Radar Physics (Our Scientific Edge): To strictly satisfy the problem statement, we computed the Circular Polarization Ratio and Degree of Polarization, successfully filtering for $CPR > 1$ and $DOP < 0.13$. However, because legacy CPR often yields false positives from rough rocks, we didn't stop there. We fed this baseline into an advanced m-chi polarimetric decomposition and a Physics-Informed Neural Network to achieve a much higher-confidence Volumetric Ice detection.

Estimated implementation cost (optional):

- **Software Licensing & R&D Cost: ₹0** We engineered the entire pipeline using open-source Python frameworks (PyTorch, SciPy) and open GIS tools (QGIS), completely eliminating the need for expensive proprietary software licenses (like MATLAB or ENVI).
- **Compute Infrastructure Budget: < ₹15,000 / month (AWS/GCP Cloud Tier)** By replacing brute-force optimization with a dense grid-scan and utilizing highly-optimized pds4_tools, our architecture requires minimal computational overhead. It can be deployed on standard ISRO server nodes without requiring expensive, dedicated GPU superclusters.
- **Data Acquisition Cost: ₹0** The architecture relies entirely on existing, freely available ISRO ISSDC Chandrayaan-2 datasets (DFSAR & OHRC) and NASA LOLA DEMs.
- **Mission Risk ROI (Return on Investment): Incalculable.** A single lunar rover mission costs hundreds of crores. By explicitly pre-calculating the exact battery Joules required via Bekker Terramechanics, LUNA-SITE mathematically prevents "dead-in-the-dark" rover loss scenarios inside the Permanently Shadowed Regions, safeguarding ISRO's entire hardware investment.

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THANK YOU